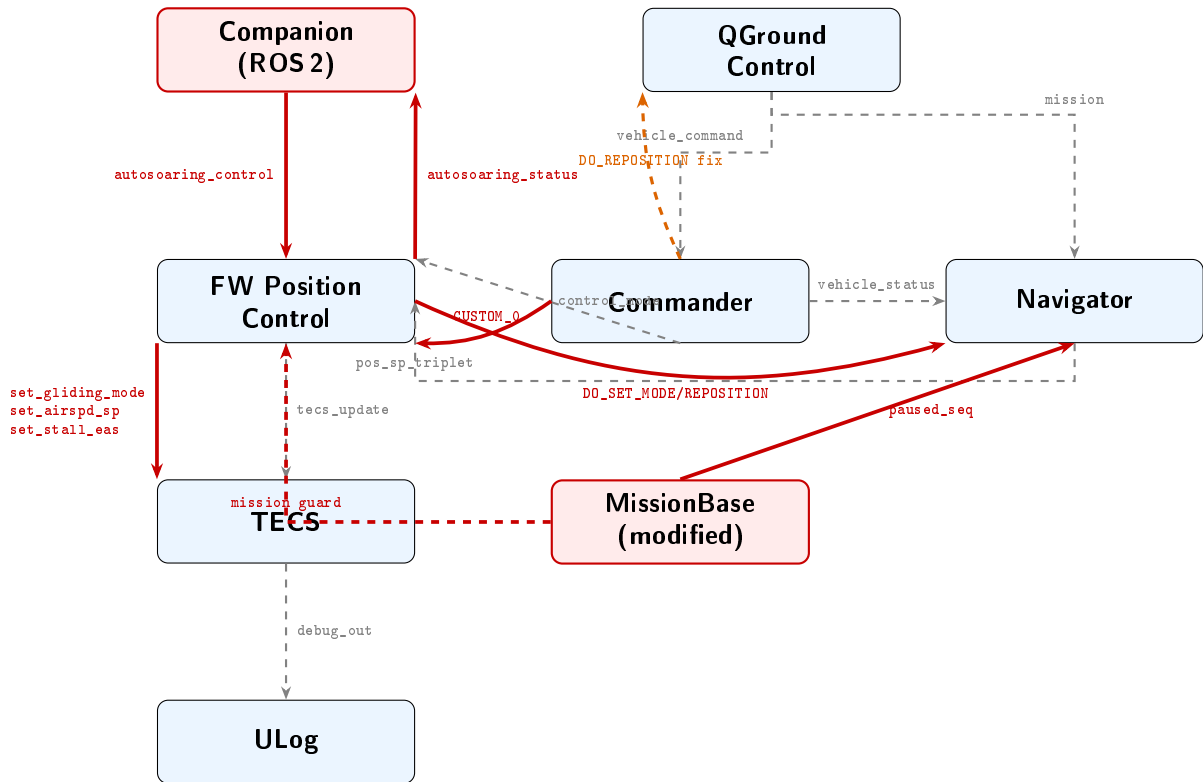


Autosoaring PX4 — Inter-Module Communication Diagram

■ — New flows - - - - Stock PX4 flows - - - - Modified flows

Radhouene

May 5, 2026

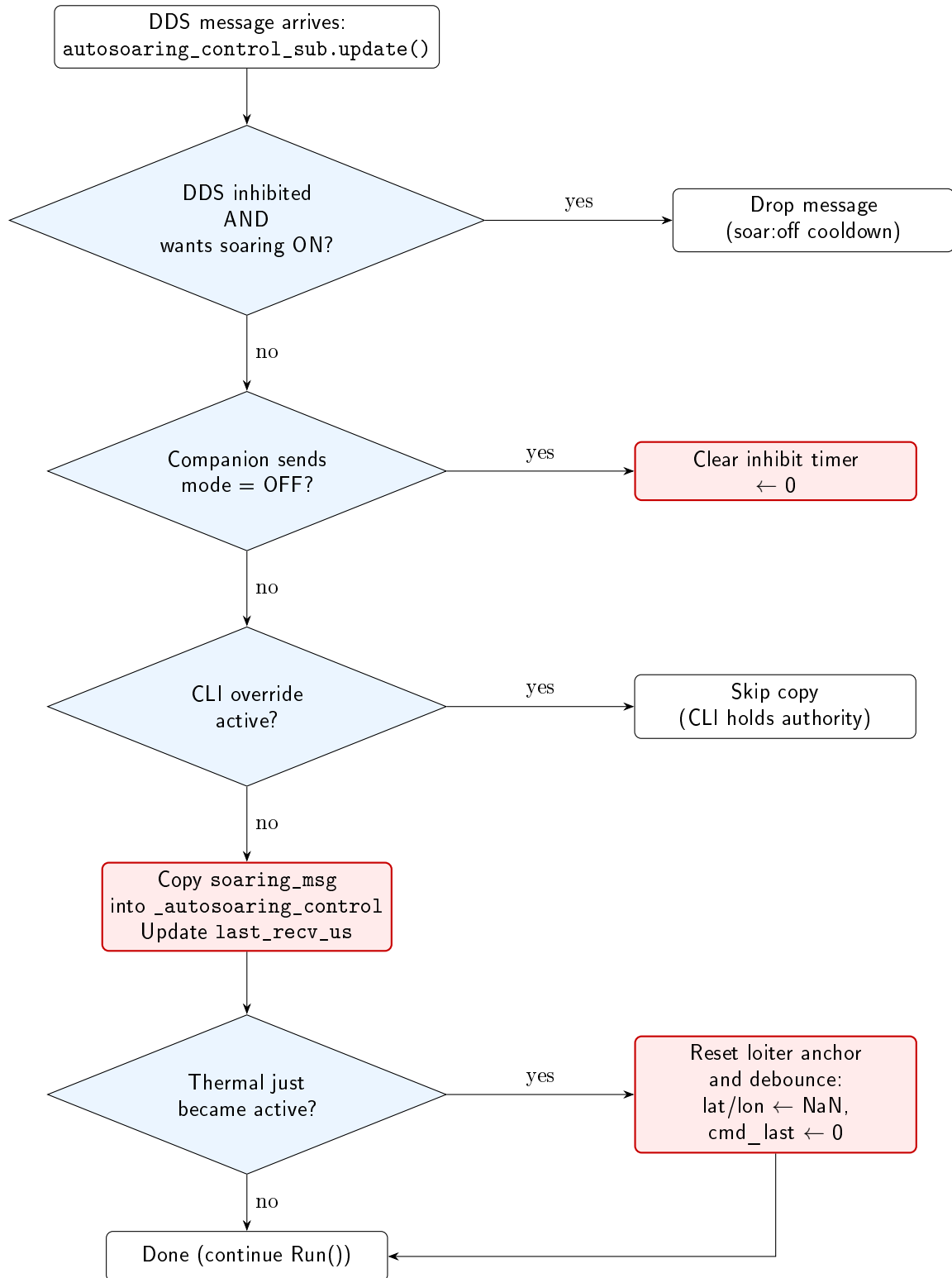


Colour	Style	Meaning
■	Solid	New flow added for autosoaring
- - - -	Dashed	Stock PX4 flow (unchanged)
- - - -	Dashed orange	Modified stock flow

Key Variables and Messages Between Modules

Variable / Topic	Publisher	Subscriber(s)	Role
<i>— New (autosoaring) —</i>			
autosoaring_control uORB	Companion ROS 2	fw_pos_control	Soaring mode, loiter centre, bank angle, airspeed cmd
autosoaring_status uORB	fw_pos_control	Companion ROS 2	FMU feedback: mode effective, DDS inhibit expiry, FMU phase
VEHICLE_CMD_CUSTOM_0	Commander (CLI)	fw_pos_control	CLI soaring mode 0–4 with optional bank/radius
VEHICLE_CMD_DO_REPOSITION	fw_pos_control	Navigator	Thermal loiter centre pin
VEHICLE_CMD_DO_SET_MODE (heartbeat)	fw_pos_control	Commander	Keep AUTO_MISSION while gliding (1 s refresh)
set_gliding_mode_enabled()	fw_pos_control	TECS	Toggle engine-off control path
set_gliding_airspeed_setpoint()	fw_pos_control	TECS	Pre-computed EAS (fixed / polar)
set_glide_stall_eas()	fw_pos_control	TECS	Glide floor from FW_AIRSPD_STALL
set_glide_i_decay()	fw_pos_control	TECS	Throttle integrator decay constant
_mission_paused_seq	MissionBase	MissionBase (internal)	Clamp stale uORB seq during AUTO_LOITER
_soaring_local_override	vehicle_command_publisher	fw_pos_control DDS poll	CLI authority flag
_soaring_dds_inhibit_until_yesh	vehicle_command_publisher	fw_pos_control DDS poll	5 s DDS cooldown after soar:off
<i>— Stock PX4 (unchanged) —</i>			
vehicle_status	Commander	Navigator, fw_pos_control	nav_state, vehicle type
vehicle_control_mode	Commander	fw_pos_control	Flags: auto, position, armed, ...
position_setpoint_triplet	Navigator	fw_pos_control	Waypoints (type, lat, lon, alt, radius)
mission	GCS / dataman	Navigator, fw_pos_control	Mission items and current seq
vehicle_command (general)	GCS/MAVLink	Commander, Navigator	Arm, calibrate, mode switch

Authority Arbitration Flow (fw_pos_control DDS Poll)



Red / orange boxes are new logic; white boxes are stock PX4 handling. The **CLI override flag** (`_soaring_local_override`) is the primary gate: when **true**, the DDS copy is skipped entirely regardless of the inhibit timer.